" How do we Blink an

Led ! "

in/kapuluru-chaithanya

MICROCHIP

PIC16F87XA

28/40/44-Pin Enhanced Flash Microcontrollers

Datasheets are given for the set of microcontrollers

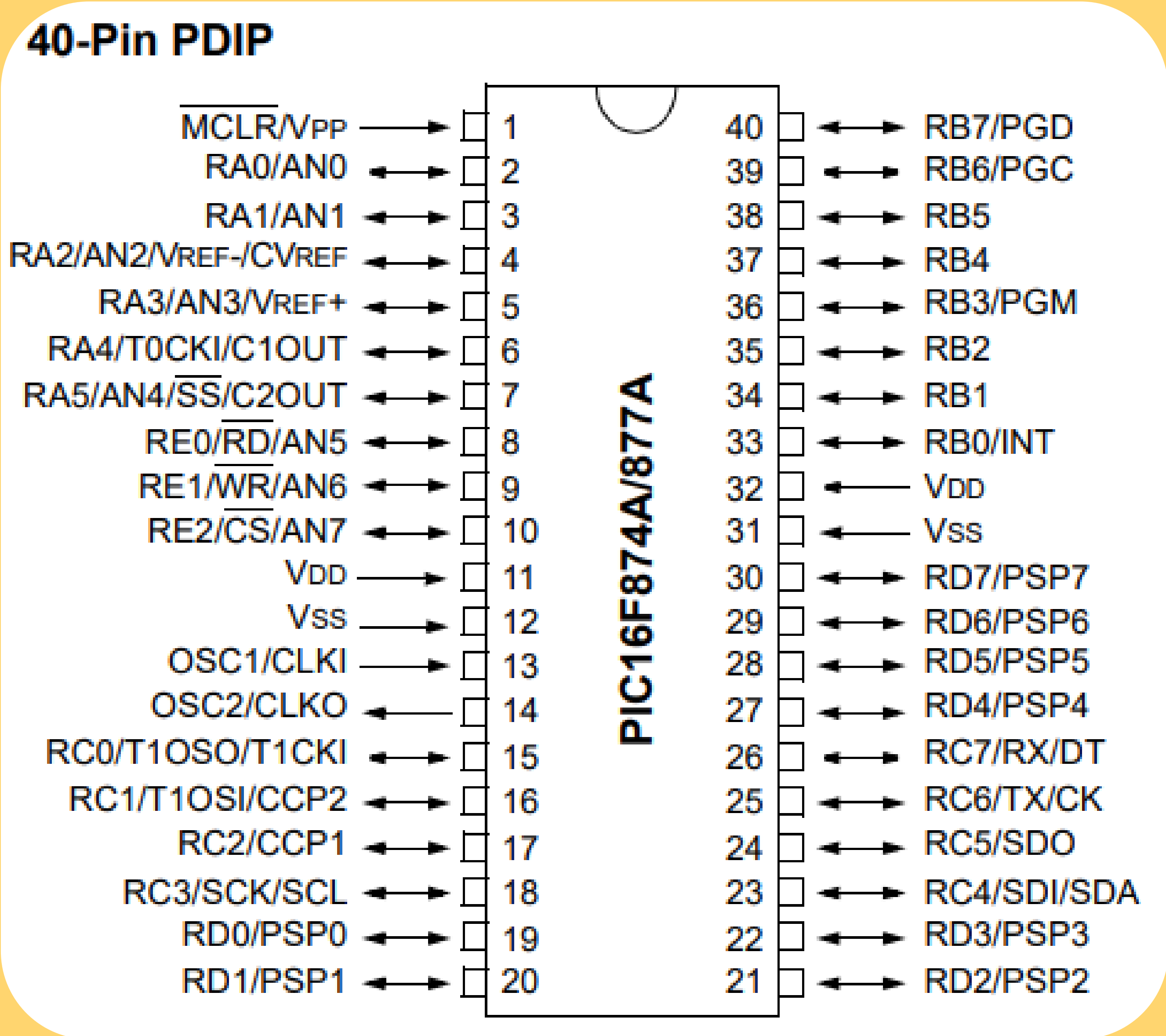
In our case, we are using PIC16F877A

Devices Included in this Data Sheet:

- PIC16F873A
- PIC16F874A
- PIC16F876A
- PIC16F877A

Steps to Blink an Led:

- Select pin / port
- Decide Input / Output
- Find the Register
- Make it high / low



- Each pin can be used for several operations

$\longrightarrow$ Towards pin Indicates Input

$\longleftarrow$ Away from pin Indicates Output

$\longleftrightarrow$ Is Bidirectional

Lets glow the led connected to Rc2

4.3 PORTC and the TRISC Register

PORTC is an 8-bit wide, bidirectional port. The corresponding data direction register is TRISC. Setting a TRISC bit (= 1) will make the corresponding PORTC pin an input (i.e., put the corresponding output driver in a High-Impedance mode). Clearing a TRISC bit (= 0) will make the corresponding PORTC pin an output (i.e., put the contents of the output latch on the selected pin).

PORTC is multiplexed with several peripheral functions (Table 4-5). PORTC pins have Schmitt Trigger input buffers.

TRISC

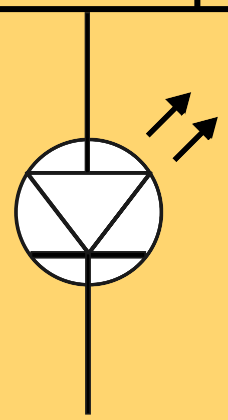
- Input → 1
- Output → 0

PORTC

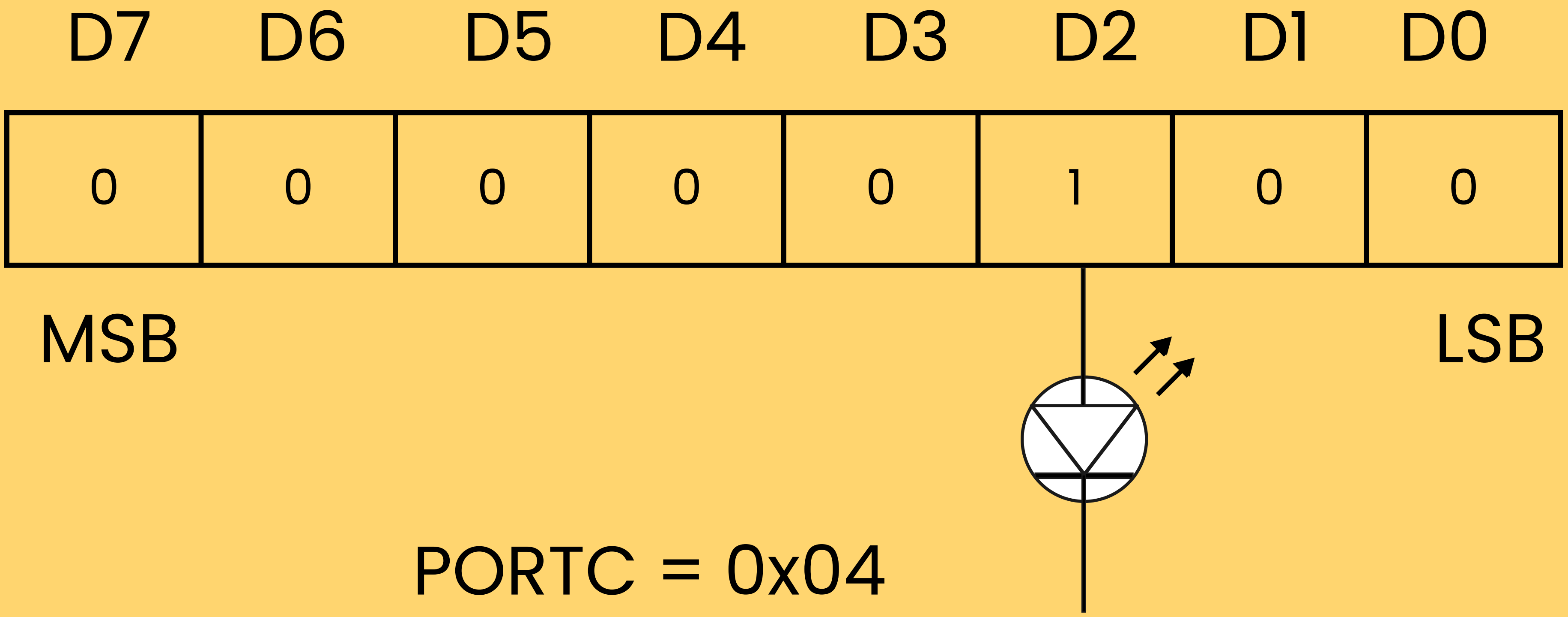
- High → 1
- Low → 0

0	0	0	0	0	0	0	0

MSB LSB



TRISC = 0x00

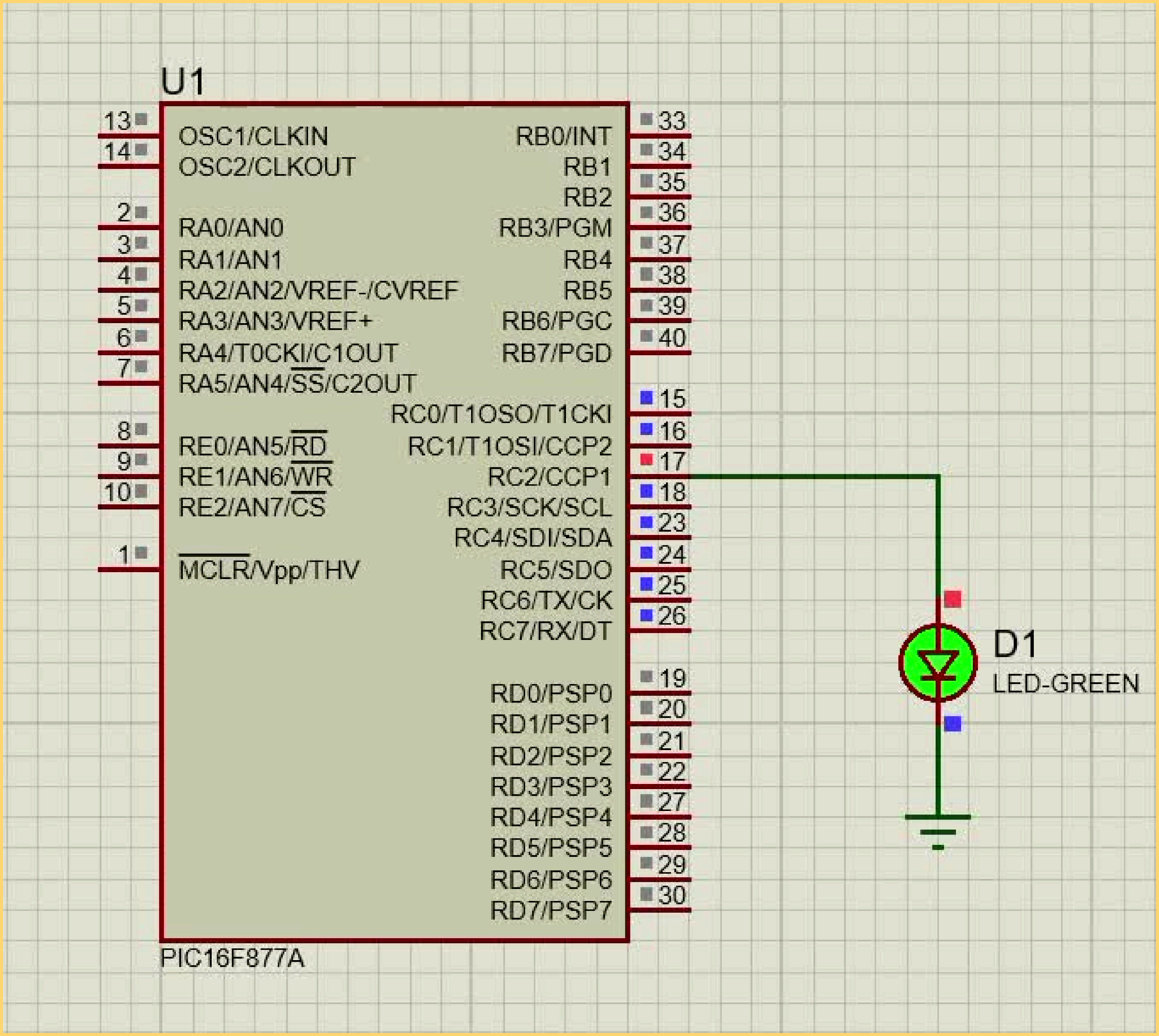


PORTC = 0x04

Dont care other bits as of now.

```c
#include <xc.h>

#define _XTAL_FREQ 6000000

void main(void) {

    TRISC = 0x00; // TRISC making all pins as output     (0000 0000)
    PORTC = 0x00; // PORTC making all pins to low state (0000 0000)

    while(1){
        PORTC = 0x04; // Making RC2 pin high (0000 0100)
        __delay_ms(1000); // 1 sec delay
        PORTC = 0x00; // Making All pins low (0000 0000)
        __delay_ms(1000);// 1 sec delay
    }

    return;
}
```

Simulation


U1
OSC1/CLKIN
OSC2/CLKOUT
RA0/AN0
RA1/AN1
RA2/AN2/VREF-/CVREF
RA3/AN3/VREF+
RA4/T0CKI/C1OUT
RA5/AN4/SS/C2OUT
RE0/AN5/RD
RE1/AN6/WR
RE2/AN7/CS
MCLR/Vpp/THV
RB0/INT
RB1
RB2
RB3/PGM
RB4
RB5
RB6/PGC
RB7/PGD
RC0/T1OSO/T1CKI
RC1/T1OSI/CCP2
RC2/CCP1
RC3/SCK/SCL
RC4/SDI/SDA
RC5/SDO
RC6/TX/CK
RC7/RX/DT
RD0/PSP0
RD1/PSP1
RD2/PSP2
RD3/PSP3
RD4/PSP4
RD5/PSP5
RD6/PSP6
RD7/PSP7
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PIC16F877A
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